

3D-Printed Normally Closed Fluidic Transistors for Soft Robots

Abstract

As a fairly new concept, soft robotics has become one of the fastest-growing fields in the robotics community. [Coined in 2008](#), “soft robotics” refers to robots constructed with soft materials, characterized by flexibility, deformability, and adaptability. However, controlling the fluid flow in these systems is still difficult, which means we need better fluidic circuit components. This paper investigates the potential of using 3D printing, specifically vat photopolymerization, to create normally-closed fluidic transistors, an essential component of a fluidic circuit. The transistor is printed in separate parts and manually assembled. Unfortunately, due to limitations in time and materials, we were not able to find concrete data on this project. However, other members of the lab have found data, which this paper will be discussing. The tests showed that without pressure applied to the gate, the transistor blocked fluid from flowing from the source to the drain (the "normally closed" state). But when we applied a gate pressure of at least 25 kPa, fluid could flow through. This work shows that 3D printing with vat photopolymerization can successfully create fluidic circuit parts through this assemble-after-printing approach, which could make soft robot technology cheaper and more accessible.

Background

As a fairly new concept, soft robotics has become one of the fastest-growing fields in the robotics community. [Coined in 2008](#), “soft robotics” refers to robots constructed with soft materials, characterized by flexibility, deformability, and adaptability. Although still in its infancy, many researchers have seen the interdisciplinary potential of soft robotics. They can be

used in a [variety of fields](#), such as healthcare, agriculture, manufacturing, exploration, rescue operations, etc.

However, soft robotics has seen significant growth primarily in the healthcare field. Thanks to the soft, flexible nature of soft robots, there has been research on potential applications of soft robotics being used in surgery. One example is its use in gastrointestinal surgery. Currently, [Imperial College London](#) is developing a self-propelled endoscopic robot that can navigate difficult conditions inside the human body, such as the gastrointestinal tract. There has also been research on [bionic prosthetics and physical therapy exosuits](#) that can provide support to patients with mobility impairments or who are recovering from injuries.

However, some major challenges that researchers have faced include actuation and cost. Currently, it takes extremely high voltages to operate soft robots. Not only will operating soft robots cost a lot of energy, but it will also be potentially hazardous in biological environments, such as surgery. As a result, researchers have shifted to using fluidic circuitry instead of electronic circuitry to operate soft robots. Not only does this help the soft robot seamlessly and safely integrate into biological environments, but it is also more cost-effective.

One such lab that researches this is the [Bioinspired Advanced Manufacturing](#) (BAM) Laboratory. However, in addition to researching fluidic circuitry, the BAM Lab also researches the application of using additive manufacturing techniques (also known as 3D printing) to build these soft robots and fluidic circuits. The use of 3D printing helps solve the problems of time, cost, and labor-intensive protocols, but also the inherent geometric (“2.5D”) constraints that traditional manufacturing techniques have. This paper investigates the potential of using 3D printing, specifically vat photopolymerization, to create normally-closed fluidic transistors, an essential component of a fluidic circuit.

Methods

The fluidic transistor was modeled using SolidWorks, a computer-aided design (CAD) software. The design is made up of three parts that are printed separately and later assembled together. These three parts are the top cap, the base structure, and the disk. When assembled together, there are four ports. One inlet for the source pressure, one outlet for the fluid to exit from, one inlet for the gate pressure, and one sealable clearing port for the bellow actuators. These parts were then separately converted into STL files and exported into Chitubox. These files were then printed on the Mars 4 LCD 3D printer using F80 Elastic Resin photomaterial.

Post-Processing

The printed parts are first soaked in isopropyl alcohol (IPA) and shaken for several minutes in a container. Then, IPA is repeatedly sent through the parts using a syringe to clear any leftover resin. The parts are then dried using N₂ gas. These steps are repeated twice to confirm that there is no residual resin.

Additive Assembly

To assemble the transistor, an additive assembly process was used. First, center the disk on the base structure such that the plug on the bottom of the disk is inside the hole on the base structure. Then, attach the top cap to the base structure such that the dovetail seal on the top cap covers the disk and fits snugly into the base structure. Secure this connection with resin and cure the added resin with UV light. Next, seal the sealable clearing port with resin and cure the added resin with UV light. Finally, add Leur Locks to the remaining three ports and secure them with UV glue. Cure the UV glue with UV light.

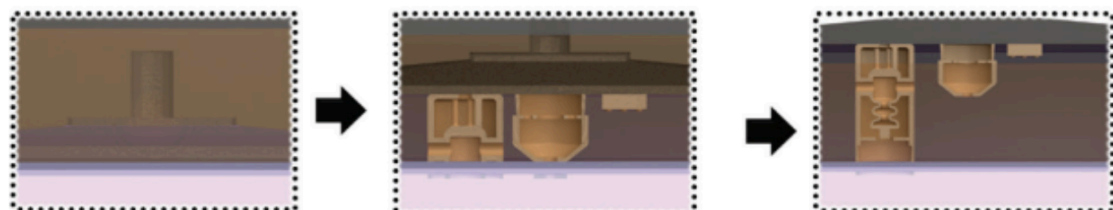
Fluidic Testing

The fluidic transistor was tested using the Fluigent FlowEZ pressure control system as well as the OxyGEN software. Dehydrated water was used as the source pressure, and air was used as the gate pressure. To test the “normally closed” mechanism of the transistor, the gate pressure was kept at 0 kPa while the source pressure was varied from 0 kPa to 75 kPa at a rate of 1 kPa/s. A working “normally closed” seal would be when there is no liquid leaking from the outlet on the base structure. To test the opening mechanism of the transistor, the source pressure was kept at a constant pressure while the gate pressure was varied from 0 kPa to 75 kPa at a rate of 1 kPa/s. A working opening mechanism would be when liquid starts coming out of the outlet on the base structure. To test the second seal, the source pressure would be constant while the gate pressure was increased until liquid no longer leaked from the outlet on the base structure.

Results

Our research team iterated over three major designs: an inverted tailed disc (with mesh), thicker bellows and disc between two poles (no mesh), and a plug disc (no mesh). Each design consisted of a base, top, and disc.

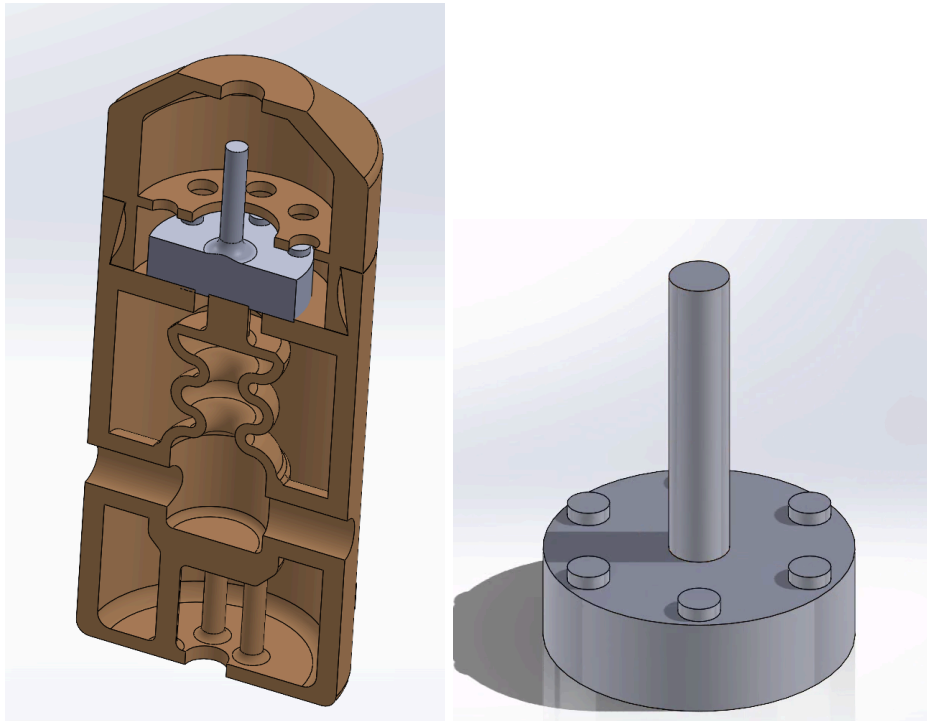
(LCD printer visual)



Changes in the base between prints were primarily to the height of the post, so as to adjust the contact between the bellows and the sealing disc, and to the bellows themselves, which

were adjusted in thickness to address printing failures and bursting caused by excessive fluid pressure. The sealing disc experienced more significant changes throughout the design process. The primary features that were printed and tested were a flat disc, a plug design, and a guide-post design. The only significant change to the top design was a mesh layer just inside the inlet, designed to prevent the disc from shifting horizontally during actuation.

Design #1:





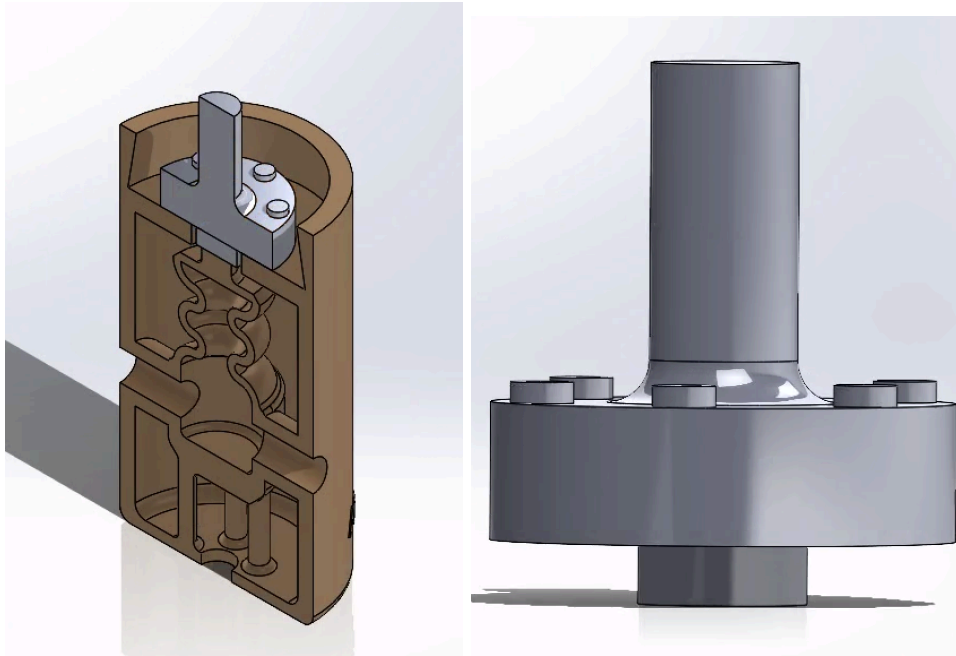
Three separate prints



Fully assembled transistor

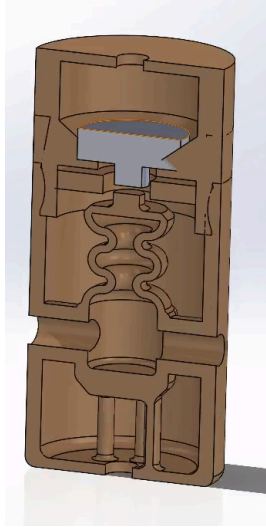
There were three components to our first design. The disk, the top half, and the bottom half. The disk had little knobs on the face of it with a long pole sticking out from the top. The top half had a slanted outer design with a mesh layer inside. The mesh layer was added to theoretically hold the disk in place by having the pole and the knobs on the disk fit snugly through the mesh layer. The bottom layer had the pathways for the gate pressure P_G to activate the bellows. When the bellows are activated, the top of the bellows pushes the disk up to allow fluid flow. This design was unsuccessful. The top half printed unreliably due to its complex design. The disk would also fail to seal reliably.

Design #2:



For the second design, the researchers increased the thickness of the guide post and disc by 1mm. This was done to limit the post's flexion in the presence of a high source pressure. To increase the disc's seal, the researchers implemented a plug design, where a 2mm extrusion of material on the underside of the disc contacts the bellows and fills the through hole while the remainder of the disc's underside continues to lie flush above the hole. The researcher also removed the mesh to limit printing failures of the top piece.

Design #3



For the third design, the researchers kept the 2mm plug on the underside of the disk and shortened the top of the bellows to compensate for the added plug. The top half of the transistor was also simplified: the mesh layer was removed, and the design was changed to a completely flat top, instead of the previous slanted design. As with the other designs, the top and the bottom halves of the transistor were joined together with a dovetail joint. The top half of the transistor was simplified due to the previous design being too complicated to print reliably. This transistor design yielded the only successful transistor, which fully sealed when the source pressure P_S overpowered the gate pressure P_G .

Despite our team lacking time to create any meaningful graphs, the group that conducted similar research previously had success. To test the normally closed transistor, experiments were conducted to measure the outflow Q by varying the source pressure P_S and the gate pressure P_G . In the first experiment, P_G was kept constant at 0 kPa, and P_S was varied from 0 to 75 kPa. It can be seen from the graph that when P_S is greater than 30 kPa, Q begins to be obstructed.

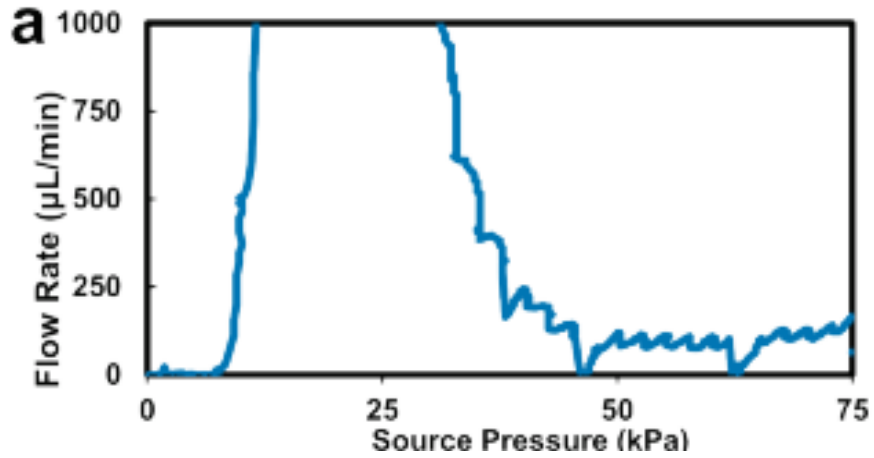


Figure A.

In the second experiment, P_s was kept at a constant 30 kPa while P_G was varied from 0 to 100 kPa. It can be seen from the graph that when P_G is greater than 20 kPa, Q begins to grow rapidly. When P_G is greater than 40 kPa, Q reaches its maximum of 1100 $\mu\text{L}/\text{min}$

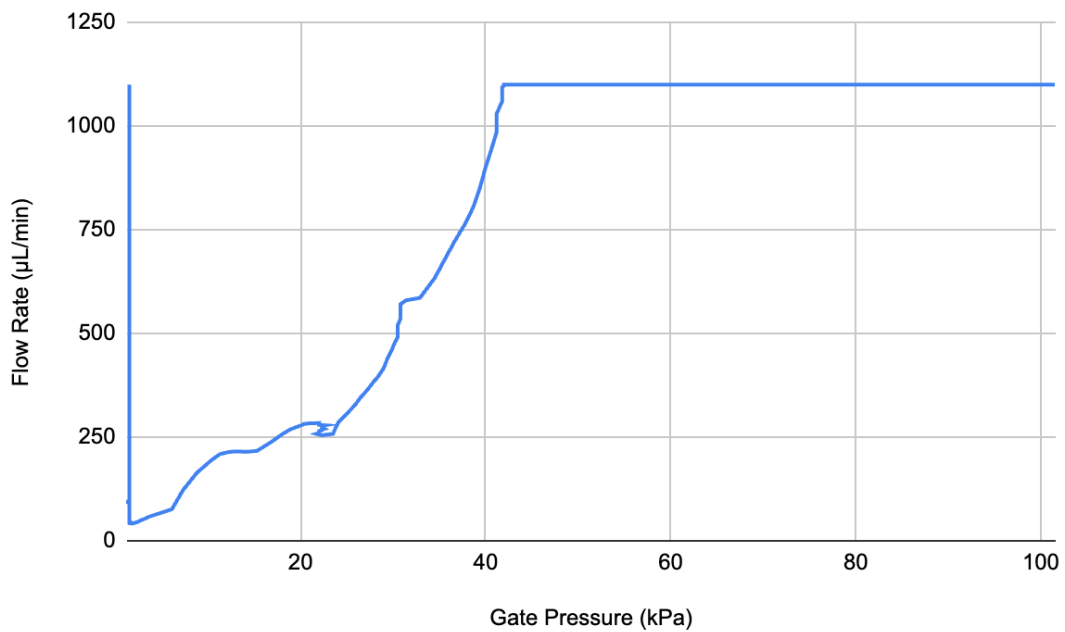


Figure B.

Discussions

Due to multiple constraints throughout the research process, there was limited data for the previously discussed designs, forcing us to use data from another group with similar designs. One of the main constraints was time. Our group only had around 6 weeks to design and test multiple prototypes. Many of the prototypes failed due to printing errors or design errors. For Design 1, the mesh layer and rounded corners of the top half of the transistor resulted in multiple print failures due to the complexity of the design. The vertical pole design for the disk also failed to seal properly. For Design 2, the mesh layer and rounded corners were removed. The vertical pole on the disk was extended to the bottom as well. However, this new disk design also resulted in printing failure due to an unreliable base to print over. To combat this, Design 3 introduced a disk with just a small pole on the bottom of the disk, removing the long vertical pole on the top. The disk was then printed upside down with the small pole facing upward. This resulted in a stable base to print on. However, due to time constraints, we were unable to collect meaningful data on Design 3.

In the first experiment, when P_G was kept at a constant pressure of 0 kPa and P_S was varied from 0 kPa to 75 kPa, it can be seen that when P_S exceeds 30 kPa ($P_S > 30$ kPa), the outflow Q begins to be inhibited. This is due to the fact that P_S begins to overpower P_G , resulting in the disk being sealed. However, in Figure A, the outflow Q can still be seen even after P_S is very large ($P_S > 50$ kPa). This could be attributed to suboptimal sealing interactions between the disk and the orifice.

In the second experiment, when P_S was kept at a constant pressure of 30 kPa and P_G was varied from 0 kPa to 75 kPa, it was observed that as P_G increased, Q also increased. This is because the disk begins to unseal more as P_G begins to overpower P_S . In addition, Figure B

shows that when PG is greater than 40 kPa, Q reaches its maximum of 1100 $\mu\text{L}/\text{min}$. What this means is that the disk has been fully unsealed, and Q can no longer increase.

Due to the limited lab space, we were only allowed to use one printer, delaying our assembly process and reducing the number of prints we produced. The type of LCD printer used was an Elegoo Mars 4, and the height of our transistors resulted in a print time of 2 hours for each design. Hence, even the smallest changes required an extended wait time. Additionally, our access to the lab depended on the graduate students' availabilities, further reducing the amount of time we had. Another limitation was the type of printer. Since our project centered around creating fluidic transistors in a cost-effective manner, we were forced to utilize a cheap printer priced between \$150 - \$200 with a low resolution. The printer also never alerted us when the lens became dirty, and hence required us to diagnose the issue by the print failures.

Towards the end of our research experience, we refined the dimensions of each component of our transistor, honing in on the best thickness and heights for our disc-plug and bellows, printing a simplistic and reliable design. The final days were spent testing with a syringe as well as our fluigent. However, due to our intensive syringe test (both with and without water for both the gate and source entryways), we ended up bursting the bellows and seeing unsuccessful tests. Despite seeing a few working iterations, our inner components lacked the strength to stay functional over repeated trials. As a result, we failed to complete a fully successful prototype. Unlike the previous research group that observed more successful seal and unseal mechanisms, these limitations held us back from attaining similar results.

Conclusion and Future Directions

All in all, although our group was not able to get formal results for our prototype, we were able to create a somewhat working prototype that worked well when testing it manually, but not with a machine. With all the limitations, such as a lack of time, print failures, and lab limitations, the fact that we were able to create this prototype is nice. Moving forward, it would be best to optimize lab time by using multiple printers instead of being limited to one. It would also be beneficial to print our designs on a more expensive and reliable printer to determine whether or not the failure was from our design or from the printer itself. In the future, we will get formal data from our own prototype to analyze. We will also move on to other fluidic circuit components, such as the normally open transistor and the diode. In the end, we will create a fully functioning hand that is powered by a 3D printed fluidic circuit, which can do a certain task, such as play the piano.

Reflections

For this project, I helped brainstorm ideas for different designs, I created our designs on SolidWorks, and I worked together with my peers when printing and postprocessing our designs. Although we all contributed equally to the final design, I was responsible for creating our designs on SolidWorks and making any needed adjustments.

I learned that research is not linear and does not have guaranteed results. There will be times when there are challenges that seem impossible to overcome. However, it is important to overcome these challenges and persevere through. Although our group was not able to create a fully working prototype, the prototype that we created was able to work somewhat well, which was much better than what we started with.

Acknowledgements

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